PCR (polymerase chain reaction)

What you're going to get / What you expect

Clean amplicons of a target DNA sequence. In this work PCR is applied to two processes: (i) amplifying the target enzyme genes from donor plasmids so they can be cloned into the pET vectors, used as a cheaper alternative to de novo gene synthesis when the sequence is already available in a repository; and (ii) colony PCR, to check that transformant colonies carry an insert of the expected size.

Materials list (per 20 uL reaction)

- Template DNA: donor plasmid (gene amplification) or a picked colony (colony PCR).
- Primers: for gene amplification, forward and reverse primers carrying NdeI (5') and HindIII (3') restriction sites; for colony PCR, universal T7 promoter and terminator primers that flank the insert.
- Pfu Ultra II high-fidelity polymerase and its buffer.
- dNTP mix.
- DMSO (to help denature GC-rich regions).
- Nuclease-free water; thermocycler; 1% (w/v) agarose gel.

Steps

1. Gene amplification from a donor plasmid: set up the reaction with 50 ng of donor plasmid as template, the NdeI/HindIII primers, dNTPs, Pfu Ultra II and DMSO, in a final volume of 20 uL.
2. Colony PCR: use a small amount of a picked colony as template and the universal T7 primers; otherwise keep the same reaction layout.
3. Run the thermocycling programme appropriate to the primer melting temperatures and amplicon length (high-fidelity polymerase).

Step	Temperature	Duration / Rate
Initial denaturation	95 °C	2 min
Amplification cycles *(12–30 cycles)*	95 °C	20–30 s
	$T_m - 5$ °C	20–30 s
	72 °C	Targets < 10 kb (vector or genomic): 1 min/kb (or 15–30 s/kb for PfuUltra II) Targets > 10 kb: 2 min/kb (or 30 s/kb at 68 °C for PfuUltra II for long targets)
Final extension	72 °C	3 min
Hold / Storage	4 °C	Hold at 4 °C (or store at -20 °C)

4. Resolve the products by 1% (w/v) agarose gel electrophoresis.
5. Compare the amplicon size with the in-silico simulation of the construct: for gene amplification, a single band of the expected size means the gene is ready for cloning; for colony PCR, a band of the expected size means the colony carries the insert.

Resources & Downloads

- NEBioCalculator: https://nebiocalculator.neb.com/
- An in silico cloning tool (e.g. SnapGene) to predict the expected amplicon sizes.

Troubleshooting

- No amplification or a weak band: reduce the amount of template (lowering the template rescued a failed amplification here), and re-check the primer annealing temperature.
- A GC-rich region will not amplify: keep DMSO in the mix and, if needed, raise the denaturation temperature or time.
- Use a high-fidelity polymerase (Pfu Ultra II) to avoid mutations in the coding sequence.